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3.8 Laplace Interpolation

In this section we look at a *missing data* or *gridding* problem, namely, how to restore missing or unmeasured values on a regular grid. Evidently some kind of interpolation from the not-missing values is required, but how shall we do this in a principled way?

One good method, already in use at the dawn of the computer age [1,2], is *Laplace interpolation*, sometimes called *Laplace/Poisson interpolation*. The general idea is to find an interpolating function y that satisfies Laplace's equation in n dimensions,

$$\nabla^2 y = 0 \quad (3.8.1)$$

wherever there is no data, and which satisfies

$$y(\mathbf{x}_i) = y_i \quad (3.8.2)$$

at all measured data points. Generically, such a function does exist. The reason for choosing Laplace's equation (among all possible partial differential equations, say) is that the solution to Laplace's equation selects, in some sense, the smoothest possible interpolant. In particular, its solution minimizes the integrated square of the gradient,

$$\int_{\Omega} |\nabla y|^2 d\Omega \quad (3.8.3)$$

where Ω denotes the n -dimensional domain of interest. This is a very general idea, and it can be applied to irregular meshes as well as to regular grids. Here, however, we consider only the latter.

For purposes of illustration (and because it is the most useful example) we further specialize to the case of two dimensions, and to the case of a Cartesian grid whose x_1 and x_2 values are evenly spaced — like a checkerboard.

In this geometry, the finite difference approximation to Laplace’s equation has a particularly simple form, one that echos the *mean value theorem* for continuous solutions of the Laplace equation: The value of the solution at any free gridpoint (i.e., not a point with a measured value) equals the average of its four Cartesian neighbors. (See §20.0.) Indeed, this already sounds a lot like interpolation.

If y_0 denotes the value at a free point, while y_u , y_d , y_l , and y_r denote the values at its up, down, left, and right neighbors, respectively, then the equation satisfied is

$$y_0 - \frac{1}{4}y_u - \frac{1}{4}y_d - \frac{1}{4}y_l - \frac{1}{4}y_r = 0 \quad (3.8.4)$$

For gridpoints with measured values, on the other hand, a different (simple) equation is satisfied,

$$y_0 = y_{0(\text{measured})} \quad (3.8.5)$$

Note that these nonzero right-hand sides are what make an inhomogeneous, and therefore generally solvable, set of linear equations.

We are not quite done, since we must provide special forms for the top, bottom, left, and right boundaries, and for the four corners. Homogeneous choices that embody “natural” boundary conditions (with no preferred function values) are

$$\begin{aligned} y_0 - \frac{1}{2}y_u - \frac{1}{2}y_d &= 0 && \text{(left and right boundaries)} \\ y_0 - \frac{1}{2}y_l - \frac{1}{2}y_r &= 0 && \text{(top and bottom boundaries)} \\ y_0 - \frac{1}{2}y_r - \frac{1}{2}y_d &= 0 && \text{(top-left corner)} \\ y_0 - \frac{1}{2}y_l - \frac{1}{2}y_d &= 0 && \text{(top-right corner)} \\ y_0 - \frac{1}{2}y_r - \frac{1}{2}y_u &= 0 && \text{(bottom-left corner)} \\ y_0 - \frac{1}{2}y_l - \frac{1}{2}y_u &= 0 && \text{(bottom-right corner)} \end{aligned} \quad (3.8.6)$$

Since every gridpoint corresponds to exactly one of the equations in (3.8.4), (3.8.5), or (3.8.6), we have exactly as many equations as there are unknowns. If the grid is M by N , then there are MN of each. This can be quite a large number; but the equations are evidently very sparse. We solve them by defining a derived class from §2.7’s `Linbcg` base class. You can readily identify all the cases of equations (3.8.4) – (3.8.6) in the code for `atimes`, below.

```
struct Laplace_interp : Linbcg {
```

[interp_laplace.h](#)

Object for interpolating missing data in a matrix by solving Laplace’s equation. Call constructor once, then solve one or more times (see text).

```
    MatDoub &mat;
    Int ii,jj;
    Int nn,iter;
    VecDoub b,y,mask;
```

```
    Laplace_interp(MatDoub_IO &matrix) : mat(matrix), ii(mat.nrows()),
    jj(mat.ncols()), nn(ii*jj), iter(0), b(nn), y(nn), mask(nn) {
```

Constructor. Values greater than $1.e99$ in the input matrix `mat` are deemed to be missing data. The matrix is not altered until `solve` is called.

```
        Int i,j,k;
        Doub vl = 0.;
```

```

for (k=0;k<nn;k++) {
    i = k/jj;
    j = k - i*jj;
    if (mat[i][j] < 1.e99) {
        b[k] = y[k] = v1 = mat[i][j];
        mask[k] = 1;
    } else {
        b[k] = 0.;
        y[k] = v1;
        mask[k] = 0;
    }
}
}

```

Fill the r.h.s. vector, the initial guess, and a mask of the missing data.

```

void asolve(VecDoub_I &b, VecDoub_O &x, const Int itrnsp);
void atimes(VecDoub_I &x, VecDoub_O &r, const Int itrnsp);

```

See definitions below. These are the real algorithmic content.

```

Doub solve(Doub tol=1.e-6, Int itmax=-1) {

```

Invoke Linbcg::solve with appropriate arguments. The default argument values will usually work, in which case this routine need be called only once. The original matrix mat is refilled with the interpolated solution.

```

    Int i,j,k;
    Doub err;
    if (itmax <= 0) itmax = 2*MAX(ii,jj);
    Linbcg::solve(b,y,1,tol,itmax,iter,err);
    for (k=0,i=0;i<ii;i++) for (j=0;j<jj;j++) mat[i][j] = y[k++];
    return err;
}
};

```

```

void Laplace_interp::asolve(VecDoub_I &b, VecDoub_O &x, const Int itrnsp) {
    Diagonal preconditioner. (Diagonal elements all unity.)
    Int i,n=b.size();
    for (i=0;i<n;i++) x[i] = b[i];
}

```

```

void Laplace_interp::atimes(VecDoub_I &x, VecDoub_O &r, const Int itrnsp) {
    Sparse matrix, and matrix transpose, multiply. This routine embodies eqs. (3.8.4), (3.8.5), and (3.8.6).

```

```

    Int i,j,k,n=r.size(),jjt,it;
    Doub del;
    for (k=0;k<n;k++) r[k] = 0.;
    for (k=0;k<n;k++) {
        i = k/jj;
        j = k - i*jj;
        if (mask[k]) {
            r[k] += x[k];
        } else if (i>0 && i<ii-1 && j>0 && j<jj-1) {
            if (itrnsp) {
                r[k] += x[k];
                del = -0.25*x[k];
                r[k-1] += del;
                r[k+1] += del;
                r[k-jj] += del;
                r[k+jj] += del;
            } else {
                r[k] = x[k] - 0.25*(x[k-1]+x[k+1]+x[k+jj]+x[k-jj]);
            }
        } else if (i>0 && i<ii-1) {
            if (itrnsp) {
                r[k] += x[k];
                del = -0.5*x[k];
                r[k-jj] += del;
            }
        }
    }
}

```

Measured point, eq. (3.8.5).

Interior point, eq. (3.8.4).

Left or right edge, eq. (3.8.6).

```

        r[k+jj] += del;
    } else {
        r[k] = x[k] - 0.5*(x[k+jj]+x[k-jj]);
    }
} else if (j>0 && j<jj-1) {
    if (itrnsp) {
        r[k] += x[k];
        del = -0.5*x[k];
        r[k-1] += del;
        r[k+1] += del;
    } else {
        r[k] = x[k] - 0.5*(x[k+1]+x[k-1]);
    }
} else {
    jjt = i==0 ? jj : -jj;
    it = j==0 ? 1 : -1;
    if (itrnsp) {
        r[k] += x[k];
        del = -0.5*x[k];
        r[k+jjt] += del;
        r[k+it] += del;
    } else {
        r[k] = x[k] - 0.5*(x[k+jjt]+x[k+it]);
    }
}
}
}

```

Top or bottom edge, eq. (3.8.6).

Corners, eq. (3.8.6).

Usage is quite simple. Just fill a matrix with function values where they are known, and with 1.e99 where they are not; send the matrix to the constructor; and call the solve routine. The missing values will be interpolated. The default arguments should serve for most cases.

```

Int m=...,n=...;
MatDoub mat(m,n);
...
Laplace_interp mylaplace(mat);
mylaplace.solve();

```

Quite decent results are obtained for smooth functions on 300×300 matrices in which a random 10% of gridpoints have known function values, with 90% interpolated. However, since compute time scales as $MN \max(M, N)$ (that is, as the cube), this is not a method to use for much larger matrices, unless you break them up into overlapping tiles. If you experience convergence difficulties, then you should call solve, with appropriate nondefault arguments, several times in succession, and look at the returned error estimate after each call returns.

3.8.1 Minimum Curvature Methods

Laplace interpolation has a tendency to yield cone-like cusps around any small islands of known data points that are surrounded by a sea of unknowns. The reason is that, in two dimensions, the solution of Laplace's equation near a point source is logarithmically singular. When the known data is spread fairly evenly (if randomly) across the grid, this is not generally a problem. *Minimum curvature methods* deal with the problem at a more fundamental level by being based on the biharmonic equation

$$\nabla(\nabla y) = 0 \quad (3.8.7)$$

instead of Laplace's equation. Solutions of the biharmonic equation minimize the integrated square of the curvature,

$$\int_{\Omega} |\nabla^2 y|^2 d\Omega \quad (3.8.8)$$

Minimum curvature methods are widely used in the earth-science community [3,4].

The references give a variety of other methods that can be used for missing data interpolation and gridding.

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